

LINEAR SYSTEMS

- SYSTEM THAT CAN BE DESCRIBED BY LINEAR EQUATIONS

LINEAR TIME INVARIANT (LTI)

- **LINEAR SYSTEM** WHERE THE RESPONSE TO A TIME SHIFTED IMPULSE IS THE SAME RESPONSE SHIFTED BY THE SAME AMOUNT

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

LINEAR DIFFERENCE EQUATIONS

- SUBSET OF **LTI SYSTEMS**
- $\sum_{k=0}^N a_k y(n-k) = \sum_{l=0}^M b_l x(n-l)$
- TYPICALLY LINEAR DIFFERENCE EQUATIONS ARE **NOT CAUSAL**
- OUTPUT IS A COMBINATION OF PAST OUTPUT AND PAST INPUT

- IF **LINEAR DIFFERENCE EQUATION** IS **CAUSAL** THEN:

$$\rightarrow y(n) = -\sum_{k=0}^N \frac{a_k}{a_0} y(n-k) + \sum_{l=0}^M \frac{b_l}{a_0} x(n-l)$$

IIR SYSTEMS

STABLE SYSTEM (BIBO)

- BOUNDED INPUT, BOUNDED OUTPUT

CAUSAL SYSTEM

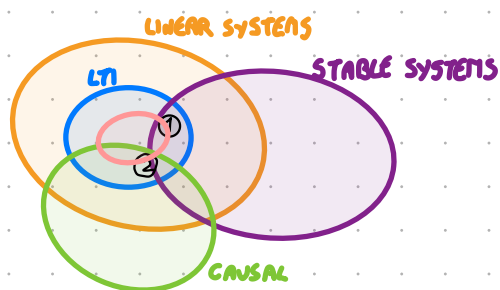
- IF A SYSTEM DOES NOT RESPOND BEFORE THE INPUT IS APPLIED
- FOR ANY n_0 , THE OUTPUT AT TIME n_0 DEPENDS ON THE INPUT ONLY FOR $n \leq n_0$
- NOT CAUSAL $\rightarrow$ OUTPUT ARRIVES BEFORE INPUT
- $x(n) = 0$ for $n < n_0 \rightarrow y(n) = 0$ for $n < n_0$

INFINITE IMPULSE RESPONSE (IIR)

- SATISFY: $y(n) = -\sum_{k=0}^N \frac{a_k}{a_0} y(n-k) + \sum_{l=0}^M \frac{b_l}{a_0} x(n-l)$
- OUTPUT DEPENDS FROM PAST OUTPUT (AND PAST INPUT)

FINITE IMPULSE RESPONSE (FIR)

- SATISFY $y(n) = \sum_{l=0}^M \frac{b_l}{a_0} x(n-l)$
- OUTPUT DOES **NOT** DEPENDS FROM PAST OUTPUT (DEPENDS ONLY PAST INPUT)



1 **LTI SYSTEM** IS **STABLE** IFF $\sum_{k=-\infty}^{\infty} |h(k)| < \infty$

2 **LTI SYSTEM** IS **CAUSAL** IFF IMPULSE RESPONSE IS 0 FOR ANY $n < 0$

